

manage to treat temperature fluctuations differently depending on the context?

Nematodes have a negative-valence neuron that triggers turning when temperatures increase, but only if the temperature is already above a certain threshold; it is a *Too hot!* neuron. Nematodes also have a *Too cold!* neuron; it triggers turning when temperatures decrease, but only when temperatures are already below a certain threshold. Together, these two negative-valence neurons enable nematodes to quickly steer away from heat when they're too hot and away from cold when they're too cold. Deep in the human brain is an ancient structure called the hypothalamus that houses temperature-sensitive neurons that work in the same way.

The Problem of Trade-Offs

Steering in the presence of multiple stimuli presented a problem: What happens if different sensory cells vote for steering in *opposite* directions? What if a nematode smells both something yummy and something dangerous at the same time?

Scientists have tested nematodes in exactly such situations. Put a bunch of nematodes on one side of a petri dish and place yummy food on the opposite side of the petri dish, then place a dangerous copper barrier (nematodes hate copper) in the middle. Nematodes then have a problem: Are they willing to cross the barrier to get to the food? Impressively, the answer is—as you would expect from an animal with even an iota of smarts—it depends. It depends on the relative concentration of the food smell versus the copper smell.

At low levels of copper, most nematodes cross the barrier; at intermediate levels of copper, only some do; at high levels of copper, no nematodes are willing to cross the barrier.

This ability to make trade-offs in the decision making process has been tested across different species of simple wormlike bilaterians and across different sensory modalities. The results consistently show that even the simplest brains, those with less than a thousand neurons, can make these trade-offs.

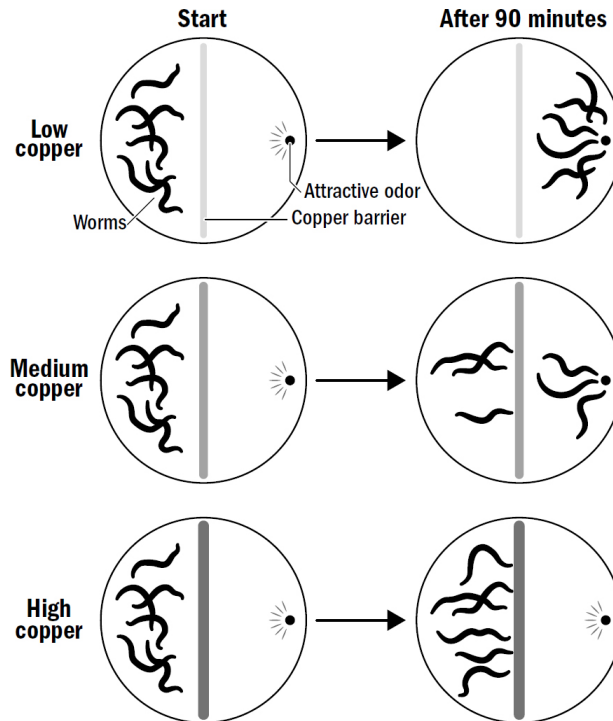


Figure 2.10
Original art by Rebecca Gelernter

This requirement of integrating input across sensory modalities was likely one reason why steering required a brain and could not have been implemented in a distributed web of reflexes like those in a coral polyp. All these sensory inputs voting for steering in different directions *had to be integrated together* in a single place to make a single decision; you can go in only one direction at a time. The first brain was this mega-integration center—one big neural circuit in which steering directions were selected.

You can get an intuition for how this works from [figure 2.9](#), which shows a simplified version of the nematode steering circuit. Positive-valence neurons connect to a neuron that triggers forward movement (what might be called a “forward neuron”), negative-valence neurons connect to a neuron that triggers turning (what might be called a “turning neuron”). The forward neuron accumulates votes for *Keep going forward!*, and the turning neuron accumulates votes for *Turn away!* The forward neuron and turn neurons mutually inhibit each other, enabling this network to integrate trade-offs and make a single choice—whichever neuron accumulates more votes wins and determines whether the animal will cross the copper barrier.

This is another example of how past innovations enabled future innovations. Just as a bilaterian cannot both go forward and turn at the same time, a coral polyp cannot both open and close its mouth at the same time. Inhibitory neurons evolved in earlier coral-like animals to enable these mutually exclusive reflexes to compete with each other so that only one reflex could be selected at a time; this same mechanism was repurposed in early bilaterians to enable them to make trade-offs in steering decisions. Instead of deciding whether to open or close one's mouth, bilaterians used inhibitory neurons to decide whether to go forward or turn.

Are You Hungry?

The valence of something depends on an animal's internal state. A nematode's choice of whether to cross a copper barrier to get to food depends not only on the relative level of food smell and copper but also on how hungry a nematode is. It won't cross the barrier to get to food if it is full, but it will if it is hungry. Further, nematodes can completely flip their preferences depending on how hungry they are. If a nematode is well fed, it will steer *away* from carbon dioxide; if hungry, it will steer *toward* it. Why? Carbon dioxide is a chemical that is released by *both* food and predators, so when a nematode is full, pursuing carbon dioxide for food isn't worth the risk of predators; when it is hungry, however, the chance that the carbon dioxide is signaling food, not predators, makes it worth the risk.

The brain's ability to rapidly flip the valence of a stimulus depending on internal states is ubiquitous. Compare the salivary ecstasy of the first bite of your favorite dinner after a long day of skipped meals to the bloated nausea of the last bite after eating yourself into a food coma. Within mere minutes, your favorite meal can transform from God's gift to mankind to something you want nowhere near you.

The mechanisms by which this occurs are relatively simple and shared across bilaterians. Animal cells release specific chemicals—"full signals" such as insulin—in response to having a healthy amount of energy. And animal cells release a different set of chemicals—hunger signals—in response to having *insufficient* amounts of energy. Both signals diffuse throughout an animal's body and provide a persistent global signal for an animal's level of hunger. The sensory neurons of nematodes have receptors that detect the presence of these signals and change their responses